JJJ-L-821A
June 4, 1969
SUPERSEDING
Fed. Spec. JJJ-L-821
October 12, 1965

FEDERAL SPECIFICATION

LITHOGRAPHIC PLATE PRESERVING GUM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE

1.1 This specification covers the requirements for lithographic plate preserving gum solution (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

JJJ-A-20 - Acacia, Technical (Gum Arabic).
PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

Federal Standards:

FED-STD-123 - Marking for Domestic Shipment (Civilian Agencies).
FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-B-26701 - Bottles, Screw Cap and Carboys, Polyethylene, Plastic.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

E 70-68 - pH of Aqueous Solutions with the Glass Electrode.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

National Motor Freight Classification Rules

(Application for copies shall be addressed to the National Classification Board, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Freight Classification Rules

(Application for copies shall be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. The gum arabic used to prepare the lithographic plate preserving gum solution shall conform to the requirements of JJJ-A-20 for class A, acacia (granular gum arabic).

3.2 Specific gravity. The gum solution shall have a specific gravity of 1.058 ± 0.005 at 15.56°/15.56°C. or a Baumé reading of 8° ± 0.5° at 60°/60° F., when tested as specified in 4.2.3 and 4.3.1

3.3 Mold inhibitor. The gum solution shall contain a preservative which the supplier knows from past experiece shall keep the gum solution in condition for use for a period of one to one and one-half years when tested as specified in 4.2.3.

3.4 Acidity (pH). The pH value of the gum solution shall be 4.3 ± 0.30 when tested as specified in 4.2.3.

3.5 Viscosity. The viscosity of the gum solution shall be 25 ± 5 seconds when tested as specified in 4.2.3 (Zahn Viscosimeter, G2 Cup).

3.6 Performance and odor. The gum solution shall produce no streaks and shall not have an unpleasant or objectionable odor, after storing for 24 to 48 hours, when tested as specified in 4.2.3 and 4.3.4. The gum solution shall protect the plate and wash up clean so that the nonprinting areas of the test plate are clean and do not accept printing ink.

3.7 Workmanship. The finished product shall be a homogeneous liquid, free from sediment and suspended matter. The occurrence of defects shall not exceed the applicable acceptable quality level (AQL) in Section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Inspection of components and materials. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.2.2 Inspection of the end item.

4.2.2.1 Examination of the end item. The end item shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth in 4.2.2.5. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled unit containers of the same capacity for the examinations in 4.2.2.2 and 4.2.2.3, in units of shipping containers for the examination in 4.2.2.4.1, or in units of pallets for the examinations in 4.2.2.4.2.

4.2.2.2 Examination for visual defects. The sample unit for this examination shall be one filled unit container.

Examine	Defects
Container	Not in accordance with contract requirements.
	Not size, shape, type or class container specified, as applicable.
	Inner or outer seal not as required.
	Inner and exterior coating, as applicable, not as specified.
	Any evidence of seepage or leakage from any part of the container including closure and seams.
	Closure not as specified.
	Bulged or distorted container.
Labeling	Not as specified, missing, incomplete, incorrect, illegible or smeared.
Contents	Not a clear, homogeneous liquid free from sediment, or suspended matter.

4.2.2.3 Examination for average net content. The sample unit for this examination shall be one filled unit container. The average net content per container shall be not less than specified.

4.2.2.4 Examination of preparation for delivery.

4.2.2.4.1 Examination for packaging, packing and marking. An examination shall be made to determine that packaging, packing and marking requirements comply with the requirements of Section 5. Defects shall be scored in accordance with the list below. The sample unit for this examination shall be one shipping container fully prepared for delivery, except that it shall not be palletized and need not be sealed. Shipping containers fully prepared for delivery that have not been palletized shall be examined for defects of closure. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defects
Marking (interior packages and shipping containers)	Omitted; incorrect; incomplete; illegible; of improper size, location, sequence, method of application, or not in accordance with contract requirements.
Packaging	Unit package not packaged as specified. Packaging material not as specified. Not in accordance with contract requirements.
Packing	Number of interior or unit packages per shipping container not as specified. Packing not as specified, closures not accomplished by specified or required methods or materials. Bulged or distorted shipping container. Not in accordance with contract requirements.
Workmanship	Inadequate application of components such as; incomplete closure of container flaps, loose or inadequate strapping or stapling.

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4.2.2.4.2 Examination for palletization. An examination shall be made to determine that palletization complies with the requirements of Section 5. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot.

Examine	Defects
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; incomplete; of improper size, location, sequence or method of application.

4.2.2.5 Inspection levels and acceptable quality levels (AQL's) for examinations. The inspection levels, for determining the sample size, and the acceptable quality levels (AQL's), expressed in defects per 100 units, shall be as follows:

Examination paragraph	Inspection levels	AQL's
4.2.2.2	I	2.5
4.2.2.3	S-2	N/A
4.2.2.4.1	S-2	2.5
4.2.2.4.2	S-1	6.5

4.2.3 Testing of the end item. The end item shall be tested for the applicable characteristics as specified in table II for each inspection lot. The lot size shall be expressed in gallons of gum solution. The sample unit for testing shall be a one gallon composite obtained by combining equal portions from samples selected at random throughout the lot in the frequency indicated in table I. No more than one sample shall be taken from a shipping container, when practicable. The number of determinations on the composite sample shall be as indicated in table II.

TABLE I. Sampling for end item testing

Lot size (gallons)	Sample frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

TABLE II. Instructions for testing of the end item

Characteristic	Specification Reference		Requirements Applicable to		Number Determinations Per Unit	Results Reported As 1/	
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest
Specific gravity	3.2	4.3.1		X	Average of 2		0.001 units or 0.1° Baumé
Mold inhibitor	3.3	2/					
Acidity (pH)	3.4	4.3.2		X	Average of 2		0.1 pH
Viscosity	3.5	4.3.3		X	Average of 2		5 secs
Performance	3.6	4.3.4		X		X	

1/ If failure is indicated, report either description of failure or the numerical point of failure, as applicable.
2/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

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4.3 Test methods.

4.3.1 Specific gravity. The specific gravity of the gum solution shall be determined by use of a hydrometer, as specified in method 4183 of FED-STD-141, except the hydrometer to be used shall be one of the following.

4.3.1.1 Specific gravity hydrometer. The range of the specific gravity hydrometer shall be 1.000 to 1.070 specific gravity units and the hydrometer scale shall be divided in 0.0005 units of specific gravity.

4.3.1.2 Baumé hydrometer. The range of the Baumé hydrometer shall be 0° to 11° Baumé and the hydrometer scale be divided in 0.1° Baumé.

4.3.2 Acidity (pH). Determine the pH value of the preserving gum solution at 25° ± 1°C. using a pH meter with a glass electrode in accordance with ASTM E 70-68.

4.3.3 Viscosity. The viscosity of the gum solution shall be determine by use of a Zahn viscosimeter, G2 cup, and taken at 72°F.

4.3.4 Performance and odor test.

4.3.4.1 Preparation of test plate. Prepare an aluminum, presensitized plate by exposing and developing it, using accepted methods.

4.3.4.2 Procedure. Place the prepared test plate on a dry table and pour a small pool (approximately 2-inches in diameter) of the sample gum solution on the center of the plate. Using a moist cellulose sponge, spread the gum over the test plate in both directions of the plate. Rub down smoothly with slightly dampened cheesecloth so that a thin, even coating of gum covers the plate. Fan the plate dry, store for 24 to 48 hours and then examine for streaks and odor. Place the plate on a printing press and wash out with a damp sponge to remove the gum. Make a short run with the plate and then inspect to ascertain whether the nonprinting areas are clean and have not accepted ink.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Gum solution shall be packaged in one quart or one gallon quantities as specified (see 6.2), in polyethylene bottles conforming to MIL-B-26701, and the bottles closed as specified therein.

5.1.2 Level C. Gum solution shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twelve 1-quart or four 1-gallon bottles of gum solution, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be provided with half-slotted fiberboard partitions that form an individual snug-fitting cell for each bottle. The partitions and the container shall be the full height of the bottles plus 1/4-inch. The partitions shall be fabricated from fiberboard conforming to type CF, class domestic, variety SW, grade 200 of PPP-F-320. Closure, waterproofing by means of tape and reinforcing shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Twelve 1-quart or four 1-gallon bottles of gum solution, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF, (variety SW), or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be provided with half-slotted partitions as specified in 5.2.1. Closure shall be in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto. The shipping container material may also be grade V4s of PPP-F-320.

5.2.3 Level C. Gum solution, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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5.3 Palletization (military requirements). Unless otherwise specified (see 6.2), gum solution packed as specified shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

5.4 Marking.

5.4.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping container shall be marked in accordance with FED-STD-123.

5.4.2 Military agencies. In addition to any special marking required by the contract or order, interior packages, shipping container and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The lithographic plate preserving gum covered by this specification is intended for use on aluminum presensitized and unsensitized plates to protect the nonimage areas from marking, abrasion, grease and to prevent plate oxidation during storage.

6.2 Ordering data. Purchaser should select the preferred options permitted and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Quantity of gum solution per package desired (see 5.1.1).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(e) When palletization is not required (see 5.3).

MILITARY INTEREST:

Custodians:

Army - GL
Air Force - 68

Review activities:

Army - GL, CE, MD, ME

User activity:

Navy - MC

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Lithographic Plate Preserving Gum JJJ-L-821A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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